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AGR 101: Principles of Agriculture - Soils, Crops and Farming Systems

The science behind food production, from soil physics to farm planning

FACULTY PRIMER

Faculty Primer

Faculty of Agriculture
Crop Science & Horticulture | 100L

Written for students, by students.

Practical notes, worked examples, practice questions and a revision plan.

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Independent study support | Not an official UNIZIK publication

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Why this pack exists

This pack introduces agriculture as applied science and business: soil formation and fertility, crop growth and management, animal production basics, farming systems and the realities of extension and markets in Nigeria.

This is a student-written study companion, not a reproduced lecture handout or a leaked examination paper. Use it to organise the topics you are taught, practise the kind of thinking your course rewards, and walk into class with better questions.

What you will be able to do

- Describe soil formation, texture, structure and fertility management.
- Explain the growth stages and management needs of a field crop.
- Compare farming systems and their suitability to Nigerian conditions.
- Identify the main constraints in Nigerian agricultural value chains.

How to use this pack

1. Skim the whole pack once in 20 minutes so you know what is inside.
2. Work through one section at a time with your lecture note and recommended text beside you.
3. Attempt the practice questions before reading the answers. Mark what you missed.
4. Use the final checklist to plan your revision week, one tick per study session.

> **Free to download, always**
> This pack is free. There is no fee, no subscription and no payment step before download, and the full PDF is not watermarked or locked. Share the link with a coursemate instead of forwarding the file.

> **Confirm your official course outline**
> Course codes, credit units and outlines can differ by department and session. Treat this pack as supplementary support and confirm the current outline with your lecturer, course rep or departmental handbook before an assessment.

Soil as a living system

Soil is the medium of plant growth, and it is a biological system, not inert dirt.

- Soil formation involves weathering, organic matter accumulation and biological activity over time.
- Texture refers to the proportions of sand, silt and clay; structure refers to how particles aggregate.
- Porosity and water-holding capacity determine irrigation and drainage needs.
- Soil pH affects nutrient availability; most crops do best in a slightly acidic to neutral range.
- Organic matter improves structure, water retention and microbial activity.
- Fertility requires both nutrients and suitable physical and biological conditions.

> **Test before you apply**
> Applying fertiliser without a soil test is guesswork. Excess nitrogen can burn crops and pollute water, while a micronutrient deficiency may go untreated.

Crop production

Crop yield is the product of genetics, environment and management.

- Land preparation: clearing, tillage and seedbed formation appropriate to the crop.
- Propagation: seed selection, nursery practice, transplanting and spacing.
- Water management: irrigation scheduling based on crop need and soil moisture.
- Weed control: mechanical, cultural, biological and chemical methods, and integrated management.
- Pest and disease management: scouting, thresholds, resistant varieties and safe pesticide use.
- Post-harvest handling determines whether the yield reaches the market in saleable condition.

Systems, livestock and value chains

A farm is a system in which crops, animals, labour and money interact.

- Shifting cultivation and bush fallow rely on long fallow periods to restore fertility.
- Mixed cropping and intercropping spread risk and use space efficiently.
- Intensive systems require higher inputs and careful management.
- Livestock production depends on breeds adapted to local conditions, feeding, housing and health.
- Extension links research to farmers but is constrained by staffing and funding.
- Value chain constraints include storage losses, poor roads, credit scarcity and price volatility.